Semantic Theory

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1 Introduction

2 The Force Model

The core interaction between two words (or nodes) i and j is

$$\mathbf{F}_{ij} = -\alpha \left(\text{cosine\_similarity}(i, j) - \beta\right) (\mathbf{x}_i - \mathbf{x}_j)$$

Where,

- cosine_similarity(i, j) measures semantic similarity between two words.
- $(\mathbf{x}_i - \mathbf{x}_j)$ is the direction vector between the two particles.
- α controls the overall force strength.
- β is a threshold that determines whether two words attract or repel each other.

2.1 Where does this come from?

This equation is inspired by several ideas from physics and machine learning.

2.1.1 Classical Physics (Hooke / Spring Systems)

The basic structure

$$\mathbf{F} \propto (\mathbf{x}_i - \mathbf{x}_j)$$

resembles a spring force, where objects pull or push along the line (or edge) connecting them.

However, instead of distance determining the force magnitude, semantic similarity determines the interaction strength.

In other words,

$$\text{distance} \longrightarrow \text{semantic similarity}$$

2.1.2 Energy-Based Models

The system can also be viewed as minimizing an implicit energy function.

- Similar words have lower energy when they are close together.
- Dissimilar words have lower energy when they are farther apart.

The force is therefore the negative gradient of the energy,

$$\mathbf{F} = -\nabla E.$$

Consequently, the simulation naturally evolves toward a semantic equilibrium.

2.1.3 Signed Interaction Field

The interaction term

$$(\text{similarity} - \beta)$$

creates two distinct regimes:

Condition	**Behavior**
similarity $> \beta$	Attraction
similarity $< \beta$	Repulsion

This balance is essential:

- Without repulsion, every particle collapses into a single point.
- Without attraction, particles simply drift apart.

The threshold parameter β therefore creates a balanced, self-organizing semantic field.